Problem Set 6
Due: Apr. 5, 2021

1. Section 3.1 Problem 2.

   Laplace random variable. Let X have the PDF

   $$f_X(x) = \frac{\lambda}{2} e^{-\lambda|x|}$$

   where λ is a positive scalar. Verify that f_X satisfies the normalization condition, and evaluate the mean and variance of X.

   Solution: We have

   $$\int_{-\infty}^{\infty} f_X(x)dx = \int_{-\infty}^{\infty} \frac{\lambda}{2} e^{-\lambda|x|}dx = 2 \cdot \frac{1}{2} \int_0^{\infty} \lambda e^{-\lambda x}dx = 2 \cdot \frac{1}{2} = 1$$

   where we have used the fact that $\int_0^\infty \lambda e^{\lambda x}dx = 1$, i.e., the normalization property of the exponential PDF. By symmetry of the PDF, we have $\mathbf{E}[X] = 0$. We also have

   $$\mathbf{E}[X^2] = \int_{-\infty}^{\infty} x^2 \frac{\lambda}{2} e^{-\lambda|x|}dx = \int_{-\infty}^{\infty} x^2 \lambda e^{-\lambda x}dx = \frac{2}{\lambda^2}$$

   where we have used the fact that the second moment of the exponential PDF is $2/\lambda^2$. Thus,

   $$\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2 = \frac{2}{\lambda^2}$$

2. Section 3.2 Problem 7.

   Alvin throws darts at a circular target of radius r and is equally likely to hit any point in the target. Let X be the distance of Alvin's hit from the center.

   (a). Find the PDF, the mean and the variance of X.

   (b). The target has an inner circle of radius t. If $X < t$, Alvin gets a score of $S = 1/X$. Otherwise his score is $S = 0$. Find the CDF of S. Is S a continuous random variable?

   Solution:

   (a). We first calculate the CDF of X. For $x \in [0, r]$, we have

   $$F_X(x) = \mathbf{P}(X \le x) = \frac{\pi x^2}{\pi r^2} = \left(\frac{x}{r}\right)^2$$

   For $x < 0$, we have $F_X(x) = 0$, and for $x > r$, we have $F_X(x) = 1$. By differentiating, we obtain the PDF

   $$f_X(x) = \begin{cases} \dfrac{2x}{r^2} & \text{if } 0 \le x \le r \\ 0 & \text{otherwise} \end{cases}$$

We have

$$\mathbf{E}[X] = \int_0^r \frac{2x^2}{r^2} dx = \frac{2r}{3}$$

$$\mathbf{E}[X^2] = \int_0^r \frac{2x^3}{r^2} dx = \frac{r^2}{2}$$

$$\text{var}(X) = E[X^2] - (E[X])^2 = \frac{r^2}{2} - \frac{4r^2}{9} = \frac{r^2}{18}$$

(b). Alvin gets a positive score in the range $[1/t, \infty)$ if and only if $X \leq t$, and otherwise he gets a score of 0. Thus, for $s < 0$, the CDF of S is $F_S(s) = 0$. For $0 \leq s < 1/t$, we have

$$F_S(s) = \mathbf{P}(S \leq s) = \mathbf{P}(\text{Alvin's hit is outside the inner circle}) = 1 - \mathbf{P}(X \leq t) = 1 - \frac{t^2}{r^2}$$

For $s \geq 1/t$, the CDF of S is given by

$$F_S(s) = \mathbf{P}(S \leq s) = \mathbf{P}(X \leq t)\mathbf{P}(S \leq s \mid X \leq t) + \mathbf{P}(X \geq t)\mathbf{P}(S \leq s \mid X \geq t)$$

We have

$$\mathbf{P}(X \leq t) = \frac{t^2}{r^2}, \quad \mathbf{P}(X > t) = 1 - \frac{t^2}{r^2}$$

and since $S = 0$ when $X > t$,

$$\mathbf{P}(S \leq s \mid X > t) = 1$$

Furthermore,

$$\mathbf{P}(S \leq s \mid X \leq t) = \mathbf{P}(1/X \leq s \mid X \leq t) = \frac{\mathbf{P}(1/s \leq X \leq t)}{\mathbf{P}(X \leq t)} = \frac{\dfrac{\pi t^2 - \pi(1/s)^2}{\pi r^2}}{\dfrac{\pi t^2}{\pi r^2}} = 1 - \frac{1}{s^2 t^2}$$

Combining the above equations, we obtain

$$\mathbf{P}(S \leq s) = \frac{t^2}{r^2} \cdot \left(1 - \frac{1}{s^2 t^2}\right) + \left(1 - \frac{t^2}{r^2}\right) \cdot 1$$

Collecting the results of the preceding calculations, the CDF of S is

$$F_S(s) = \begin{cases} 0 & \text{if } s < 0 \\ 1 - \dfrac{t^2}{r^2} & \text{if } 0 \leq s < 1/t \\ 1 - \dfrac{1}{s^2 r^2} & \text{if } s \geq 1/t \end{cases}$$

Because F_S has a discontinuity at $s = 0$, the random variable S is not continuous.

3. An image is partitioned into two regions, one white and the other black. A reading taken from a randomly chosen point in the white section will be normally distributed with $\mu = 4$ and $\sigma^2 = 4$, whereas one taken from a randomly chosen point in the black region will have a normally distributed reading with parameters $(6, 9)$. A point is randomly chosen on the image and has

a reading of 5. If the fraction of the image that is black is α, for what value of α would the probability of making an error be the same, regardless of whether one concluded that the point was in the black region or in the white region?

Solution: Applying Bayes' rule,

$$\begin{aligned}
&\mathbf{P}(\text{the point is in the black region} \mid \text{the reading is 5})\\
=&\frac{\mathbf{P}(\text{black and 5})}{\mathbf{P}(5)}\\
=&\frac{\mathbf{P}(5 \mid \text{black})\mathbf{P}(\text{black})}{\mathbf{P}(5 \mid \text{black})\mathbf{P}(\text{black}) + \mathbf{P}(5 \mid \text{white})\mathbf{P}(\text{white})}\\
=&\frac{\frac{1}{3\sqrt{2\pi}}e^{-(5-6)^2/(2\cdot 9)} \cdot \alpha}{\frac{1}{3\sqrt{2\pi}}e^{-(5-6)^2/(2\cdot 9)} \cdot \alpha + \frac{1}{2\sqrt{2\pi}}e^{-(5-4)^2/(2\cdot 4)} \cdot (1-\alpha)}\\
=&\frac{\frac{\alpha}{3}e^{-1/18}}{\frac{\alpha}{3}e^{-1/18} + \frac{1-\alpha}{2}e^{-1/8}}
\end{aligned}$$

α is the value that makes the preceding equal 1/2. Therefore,

$$\alpha = \frac{3e^{-1/8}}{3e^{-1/8} + 2e^{-1/18}} \approx 0.583$$

4. Show that $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

   Hint: $\Gamma(t) = \int_0^\infty e^{-x}x^{t-1}dx$. Make the change of variables and then relate the resulting expression to the normal distribution.

   Solution:

$$\begin{aligned}
\Gamma(\frac{1}{2}) &= \int_0^\infty e^{-x}x^{-1/2}dx\\
&= \sqrt{2}\int_0^\infty e^{-y^2/2}dy \qquad \text{by } x = y^2/2\\
&= 2\sqrt{\pi}\int_0^\infty \frac{1}{\sqrt{2\pi}}e^{-y^2/2}dy\\
&= 2\sqrt{\pi} \cdot \frac{1}{2}\\
&= \sqrt{\pi}
\end{aligned}$$